

## **PHYSIOLOGY**

- 1- **What is the effect of parasympathetic nervous system activation on airways?**
  - A- Increasing of anatomic dead space
  - B- Bronchodilation
  - C- **Bronchoconstriction**
  - D- Histamine release and allergic reaction
  
- 2- **Which of the following occur during normal inspiration?**
  - A- **Diaphragm contraction**
  - B- Diaphragm relaxation
  - C- Contraction of internal intercostal muscles
  - D- Relaxation of internal intercostal muscles
  
- 3- **which one the following can't be directly measured by spirometry ?**
  - A- Tidal volume
  - B- **Residual Volume**
  - C- Inspiratory Reserve Volume
  - D- Expiratory Reserve Volume
  
- 4- **Which gas is used for estimating of “diffusing capacity of O<sub>2</sub>” in the lung?**
  - A- **CO**
  - B- O<sub>2</sub>
  - C- CO<sub>2</sub>
  - D- Helium
  
- 5- **Which one of the following muscles has a role in deep expiration?**
  - A- scalene
  - B- anterior serrati
  - C- sternocleidomastoid muscles
  - D- **the abdominal recti**
  
- 6- **What is the amount of Transpulmonary pressure at the beginning of normal inspiration?**
  - A- -7.5 centimeters of water
  - B- +7.5 centimeters of water
  - C- -5 centimeters of water
  - D- **+5 centimeters of water**

- 7- **What is the effect of surfactant on lung compliance?**  
A) It depends on the amount of pleural pressure  
B) Leads to a decrease  
C) Leads to an increase  
D) Has no effect
- 8- **What is the “Residual volume”?**  
A) The volume of air remaining in the lungs after the most forceful expiration  
B) The maximum amount of air a person can expel from the lungs after first filling the lungs to their maximum extent  
C) The amount of air that remains in the lungs at the end of normal expiration  
D) The maximum extra volume of air that can be expired by forceful expiration
- 9- **Which zone of the lung *has* blood flow during both systole and diastole phases in standing position?**  
A) Zone-I  
B) Zone-II  
C) Zone-III  
D) Zone-IV
- 10- **Which one the following decreases the affinity of hemoglobin for O<sub>2</sub>?**  
A- An increase in pH  
B- A decrease in CO<sub>2</sub> pressure  
C- A decrease in DPG concentration  
D- An increase in body temperature
- 11- **What is the “anatomical shunt”?**  
A) Decreasing ventilation/perfusion ratio ( $V_A/Q$ ) to zero  
B) Mixture of pulmonary vein blood and bronchial vein blood  
C) Increasing of  $V_A/Q$  ratio 2.5 times more than normal amount  
D) Mixture of pulmonary arterial and venous blood
- 12- **What changes are happened in the breathing pattern by increasing the activity of the pneumotaxic center?**  
A- Inspiration and expiration become deeper  
B- The breathing rate increases  
C- The tidal volume increases  
D- It doesn't make a change

**13- Which one is correct about ventilation and perfusion in the lungs?**

- A- The blood flow decreases from the top of the lung to bottom
- B- The ventilation to perfusion ratio (V/Q) increases from top of the lung to bottom
- C- The ventilation increases from the top of the lung to the bottom**
- D- V/Q ratio is same for all levels of the lung in upright person

**14- What is the normal value of CO<sub>2</sub> pressure in systemic arterial blood?**

- A- 104 mmHg
- B- 95 mmHg
- C- 45 mmHg
- D- 40 mmHg**

**15- What is the maximum amount of CO<sub>2</sub> transportation in the blood?**

- A) As bicarbonate**
- B) Dissolved in the blood.
- C) Bound to the hemoglobin.
- D) As carboxyhemoglobin.

**16- What is the normal value of utilization coefficient for O<sub>2</sub> in a normal condition?**

- A- 20%
- B- 25%**
- C- 30%
- D- 35%

**17- Which one is correct about the role of the dorsal respiratory group in medulla?**

- A- Become active during deep expiration
- B- Inhibits the pneumotaxic center in pons
- C- Become active during deep inspiration
- D- Sends its action potential to phrenic nerves**

**18- In a patient with pulmonary fibrosis, for a pressure difference of 4 mm Hg between the two sides of the respiratory membrane, the amount of 400 ml of CO<sub>2</sub> is exchanged per minute, what will be the diffusing capacity of this gas?**

- A- 25 ml/min/mmHg
- B- 50 ml/min/mmHg
- C- 100 ml/min/mmHg**
- D- 200 ml/min/mmHg

**19- Which gas is used for estimating of “diffusing capacity of O<sub>2</sub>” in the lung?**

- a. CO<sub>2</sub>
- b. CO
- c. O<sub>2</sub>
- d. He

**20- What is the normal pleural pressure at the beginning and end of normal expiration, respectively?**

- A- -7.5 and -5 cmH<sub>2</sub>O
- B- -7.5 and 0 cmH<sub>2</sub>O
- C- -5 and -7.5 cmH<sub>2</sub>O
- D- -5 and 0 cmH<sub>2</sub>O

**21- How much cilia are there on each epithelial cell in the mucociliary system in airways?**

- A- 20
- B- 200
- C- 2000
- D- 20000

**22- Which one of the following respiratory works is less than others?**

- A- Compliance work
- B- Airway resistance work
- C- Tissue resistance work
- D- All the works are equal

## **HISTOLOGY AND EMBRIOLOGY**

**23- Where are there club cells?**

- A) Lamina propria
- B) trachea
- C) bronchiole
- D) bronchus

**24- Which type of these cells is the most or least in microscopic structure of the respiratory epithelium?**

- A) Ciliated columnar- Kolchitsky
- B) Goblet- cuboidal
- C) Brush border- DNES
- D) Supporting- Stem cells

**25- In the histological structure of the trachea.....**

- A) All epithelial cells have acidophilic granules.
- B) The C-shaped cartilage covered by perichondrium.
- C) Serous glands in the lamina propria produce mucus.
- D) Hyaline C-shaped cartilage is lined by epithelium.

**26- Which of these structures are lined by 2 types of epithelia?**

- A. Trachea
- B. Bronchus
- C. Olfactory zone
- D. Epiglottis of larynx

**27- Which of these cells have a defense role in lung?**

- A. Pneumococcus Type I
- B. Pneumococcus Type II
- C. Pneumococcus Type III
- D. Club cells

**28- Which one of the following statements is TRUE about the Canalicular period?**

**A- The alveolar ducts will start forming at this period.**

B- Prematurely born fetus can survive.

C- The epithelium of the terminal bronchi becomes very thick.

D- Capillaries begin to bulge at this period.

**29- Which of the following linings does the laryngotracheal diverticulum give rise to the muscles & Glands of the respiratory system, respectively?**

A. The ectoderm - The Mesoderm

**B. The Mesoderm- Laryngeal Endoderm**

C. The endoderm - Laryngeal arches

D. The mesoderm of 5th and 6th pair of pharyngeal arches

## **ANATOMY**

**30- Bifurcation of trachea is at the level of ....**

**A- Intervertebral disc between T4 and T5**

B- Third costal cartilage

C- Xiphoid process of sternum

5<sup>th</sup> intercostal space

**31- Which muscle is innervated by external laryngeal nerve?**

A- Transverse arytenoid

B- Posterior cricoarytenoid

C- Vocalis

**D- Cricothyroid**

**32- Where is the inferior border of the lung in the mid-clavicular line?**

**A- 6<sup>th</sup> rib**

B- 8<sup>th</sup> rib

C- 10<sup>th</sup> rib

D- 12<sup>th</sup> rib

**33- Maxillary sinus drains into the ...**

- A- Superior meatus
- B- Middle meatus**
- C- Inferior meatus
- D- Sphenoethmoidal recess

**34- Which one is a branch of the maxillary artery?**

- A- Superior labial a.
- B- Sphenopalatine a.**
- C- Anterior ethmoidal a.
- D- Lateral nasal a.

**35- Galen anastomosis under piriform recess is formed by ...**

- A- Internal laryngeal and external laryngeal nerves
- B- Internal laryngeal and recurrent laryngeal nerves**
- C- External laryngeal and recurrent laryngeal nerves
- D- Recurrent laryngeal and pharyngeal nerves

**36- Which nerve does supply sensory innervation of the larynx from laryngeal inlet to vocal folds?**

- A- Recurrent laryngeal n.
- B- Internal laryngeal n.**
- C- External laryngeal n.
- D- Inferior laryngeal n.

**37- Mediastinal pleura is innervated by ...**

- A- Intercostal nerves
- B- Vagus nerve
- C- Sympathetic trunk
- D- Phrenic nerve**

**38- Which bone on the lateral wall of the nasal cavity is a separate bone of the facial skeleton?**

- A- Nasal bone
- B- Ethmoid
- C- Vomer
- D- Inferior nasal concha**

**39- Where is the location of the palatine tonsil?**

- A- Roof of the nasopharynx
- B- Lateral wall of the oropharynx**
- C- Lateral wall of the nasopharynx
- D- Anterior wall of the laryngopharynx

**40- Which one is formed by Inferior free border of the quadrangular membrane?**

- A- Vestibular ligament**
- B- Vocal ligament
- C- Cricothyroid ligament
- D- Thyrohyoid ligament